

Credit Card Approval Prediction Using Machine Learning Project

Description:

The **Credit Card Approval Prediction** system is a machine learning-based web application that predicts whether a credit card application is likely to be approved or rejected based on an applicant's personal and financial information. The system analyzes features such as age, annual income, employment status, credit score, loan history, and other financial details to assist financial institutions in making faster and more accurate decisions.

The application is developed using **Python, Flask, Scikit-learn, HTML, and CSS**, and is deployed on **PythonAnywhere**. The trained **Random Forest Classifier** model provides real-time predictions through a user-friendly web interface.

Scenarios

Scenario 1: High Credit Score Applicant

A user enters applicant details including a high credit score, stable employment, and sufficient annual income. The system processes the information and predicts **Approved**, indicating a high probability of credit card approval.

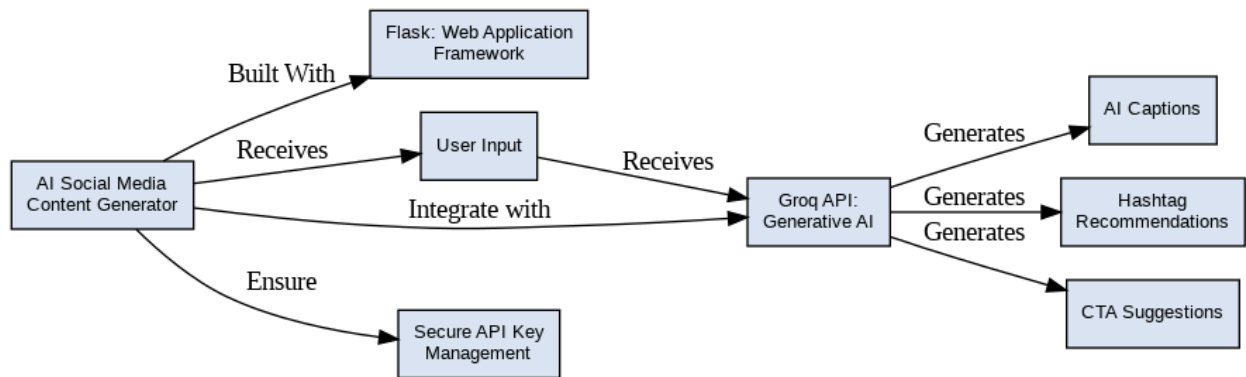
Scenario 2: Low Credit Score Applicant

A user enters details with a low credit score, unstable employment, and high debt. The machine learning model predicts **Rejected**, helping identify high-risk applicants.

Scenario 3: Borderline Applicant

A user provides moderate financial information with an average credit score. The system analyzes all features and predicts the approval status based on the trained Random Forest model.

Technical Architecture:



Pre-requisites:

- ☐ Python 3.10 or above
- ☐ Flask
- ☐ Pandas
- ☐ NumPy
- ☐ Scikit-learn
- ☐ Joblib
- ☐ HTML & CSS
- ☐ Git and GitHub
- ☐ PythonAnywhere account

Project Workflow:

Milestone 1 – Data Collection and Model Selection

Activity 1.1

Collect the Credit Card Approval dataset.

Activity 1.2

Preprocess the dataset by handling missing values and encoding categorical features.

Activity 1.3

Select the Random Forest Classifier.

Activity 1.4

Split the dataset into training and testing data.

Milestone 2 – Model Development

Activity 2.1

Train the Random Forest model.

Activity 2.2

Evaluate the model accuracy.

Activity 2.3

Save the trained model using Joblib.

Milestone 3 – Flask Application Development

Activity 3.1

Develop app.py.

Activity 3.2

Load

- model.pkl
- scaler.pkl
- label_encoders.pkl

Activity 3.3

Accept user inputs from the HTML form.

Activity 3.4

Predict approval status.

Activity 3.5

Display prediction result.

Milestone 4 – Frontend Development**Activity 4.1**

Develop responsive HTML pages.

Activity 4.2

Create CSS styling.

Activity 4.3

Display prediction results.

Milestone 5 – Deployment**Activity 5.1**

Deploy the Flask application on PythonAnywhere.

Activity 5.2

Test the deployed application.

Activity 5.3

Publish the live URL.

Application Pages**Home Page**

The home page provides a form where users enter applicant details such as:

- Gender

- Age
- Annual Income
- Employment Status
- Credit Score
- Number of Loans
- Total Debt
- Bank Balance
- Credit Card Count
- Other financial details

Users click the **Predict** button to submit the application.

Credit Card Approval Prediction

Gender (1=Male, 0=Female)

Male

Age

Debt

Married (1=Yes, 0=No)

Yes

Bank Customer (1=Yes, 0=No)

Yes

Industry

CommunicationServices

Ethnicity

Asian

Years Employed

Years Employed

Prior Default (1=Yes, 0=No)

No ▼

Employed (1=Yes, 0=No)

Yes ▼

Credit Score

Drivers License (1=Yes, 0=No)

Yes ▼

Citizen

ByBirth ▼

Zip Code

Income

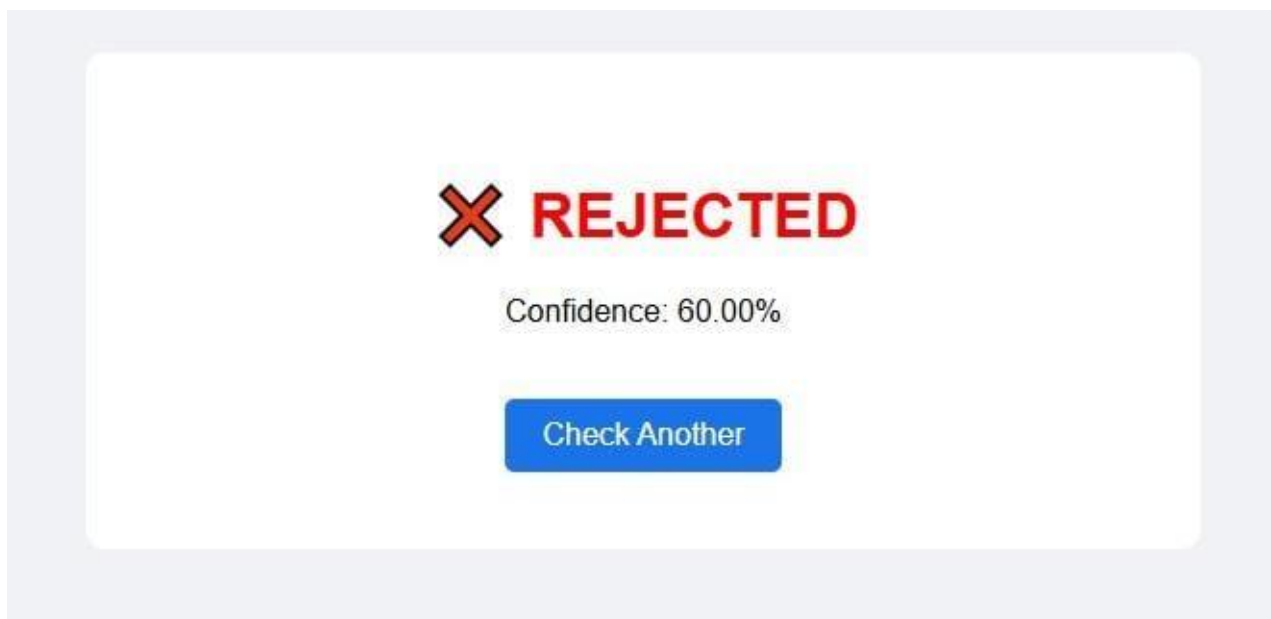
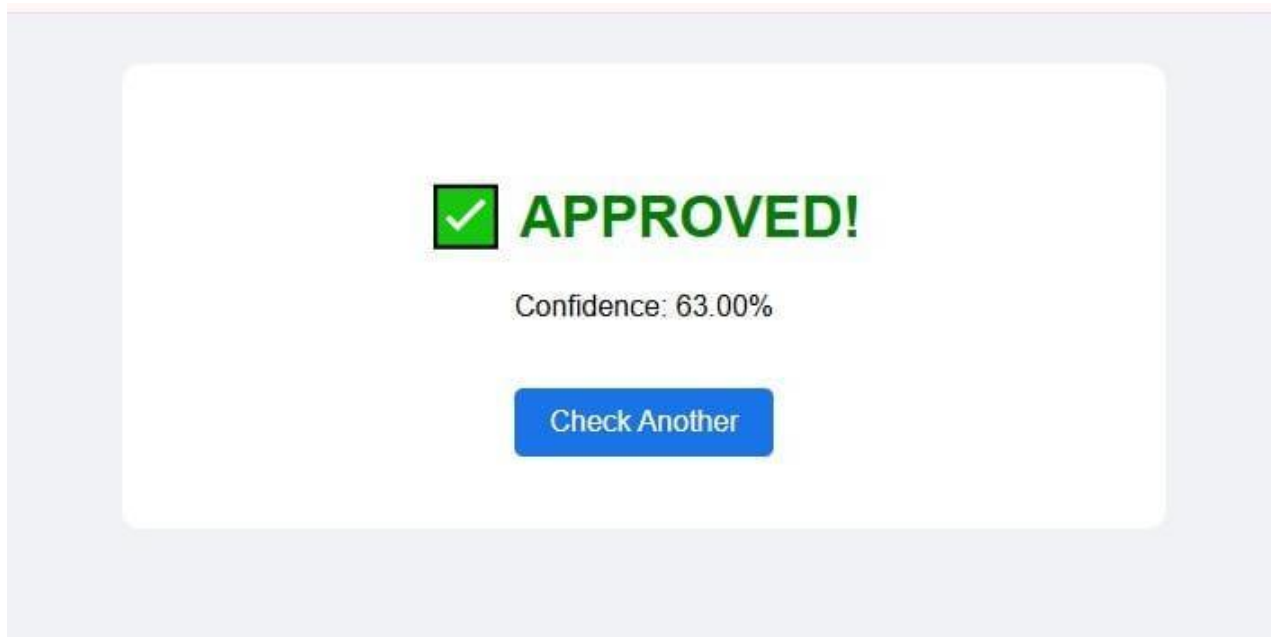
Predict Approval

Prediction Result Page

The application displays:

- Applicant Details
- Prediction Status (Approved/Rejected)
- Prediction Confidence
- Approval Probability

- Rejection Probability



Deployment

Hosting Platform: PythonAnywhere

Live URL:

<https://yojana4060.pythonanywhere.com/>

Conclusion

The Credit Card Approval Prediction project demonstrates how machine learning can automate the credit card approval process by analyzing applicant information and predicting approval status with high accuracy. The application provides a simple web interface built using Flask, HTML, and CSS, enabling users to obtain real-time predictions. By reducing manual effort and improving decision-making speed, the system offers a practical solution for financial institutions. Future enhancements may include database integration, user authentication, advanced analytics dashboards, and support for additional machine learning algorithms.

